

- (ii) Wat is die versnelling  $a$  van die massa  $m$  in terme van onder andere die spankrag  $T$  in die tou.

*What is the acceleration  $a$  of the mass  $m$  in terms of the tension  $T$  in the rope.*

$$F = ma \quad \tau = I\alpha$$

$$= TR \quad a = \frac{TR}{m}$$

(2)

- (iii) Toon aan dat  $T = \frac{mg}{1 + \frac{mR^2}{I}}$   $\alpha = \frac{TR}{I}$   $T = mg - ma$
- Show that  $T = \frac{mg}{1 + \frac{mR^2}{I}}$   $\tau = I\alpha = TR$   $T = \frac{I\alpha}{R}$

$$T = mg - ma$$

$$T = \frac{1}{2} mR^2 \alpha$$

(3)

- (iv) Die traagheidsmoment vir 'n soliede wiel word gegee deur  $I = \frac{1}{2}MR^2$ . Indien  $I = 0,09 \text{ kg m}^2$  is vir 'n wiel met 'n deursnit van 60 cm. bepaal sy massa.

*The moment of inertia of a solid wheel is given by  $I = \frac{1}{2}MR^2$ . If  $I = 0,09 \text{ kg m}^2$  for a wheel with a diameter of 60 cm, determine the mass of the wheel.*

$$M = \frac{2I}{R^2}$$

$$= 0,5 \text{ kg}$$

(2)

- (v) Met 'n massa  $m = 0,5 \text{ kg}$  wat aan die tou hang, bepaal die spankrag in die tou.

*With a mass  $m = 0,5 \text{ kg}$  attached to the wheel, determine the tension in the rope.*

$$1,62 \text{ N}$$

(3)

- (vi) Bereken die hoekversnelling  $\alpha$  van die wiel.

*Determine the angular acceleration  $\alpha$  of the wheel.*

$$\alpha = \frac{TR}{I}$$

$$= 10,9 \text{ rad/s}^2$$

(3)